

Presentation on

Automatic Power Factor Correction System

By

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- High electrical loads can lead to a poor power factor, resulting in higher power losses and increased energy costs. An Automatic Power Factor Correction (APFC) system helps improve electrical efficiency by continuously monitoring and maintaining the power factor within an acceptable range.
- A low power factor can result in:
 - Higher power losses
 - Increased electricity bills
 - Additional penalties from electricity utilities
- To address these issues, this project proposes an **STM32-based Automatic Power Factor Correction system** designed to improve power factor for both household loads and high-power industrial applications.

Problem Statement

- High electrical load systems often operate with poor power factor due to the predominance of inductive loads. This results in increased reactive power demand, higher power losses, reduced system efficiency, and increased electricity costs. Conventional power factor correction methods are either manual or lack real-time monitoring and adaptability to varying load conditions.

Objectives :

- To enhance power factor close to 0.95 to 1 while minimizing reactive power losses.
- To monitor real-time energy consumption for effective load management.
- To reduce electricity cost and demand penalties through automatic control.
- To design a Smart APFC system that reduces harmonics generated due to frequent capacitor bank switching during load variations.

Proposed Block Diagram

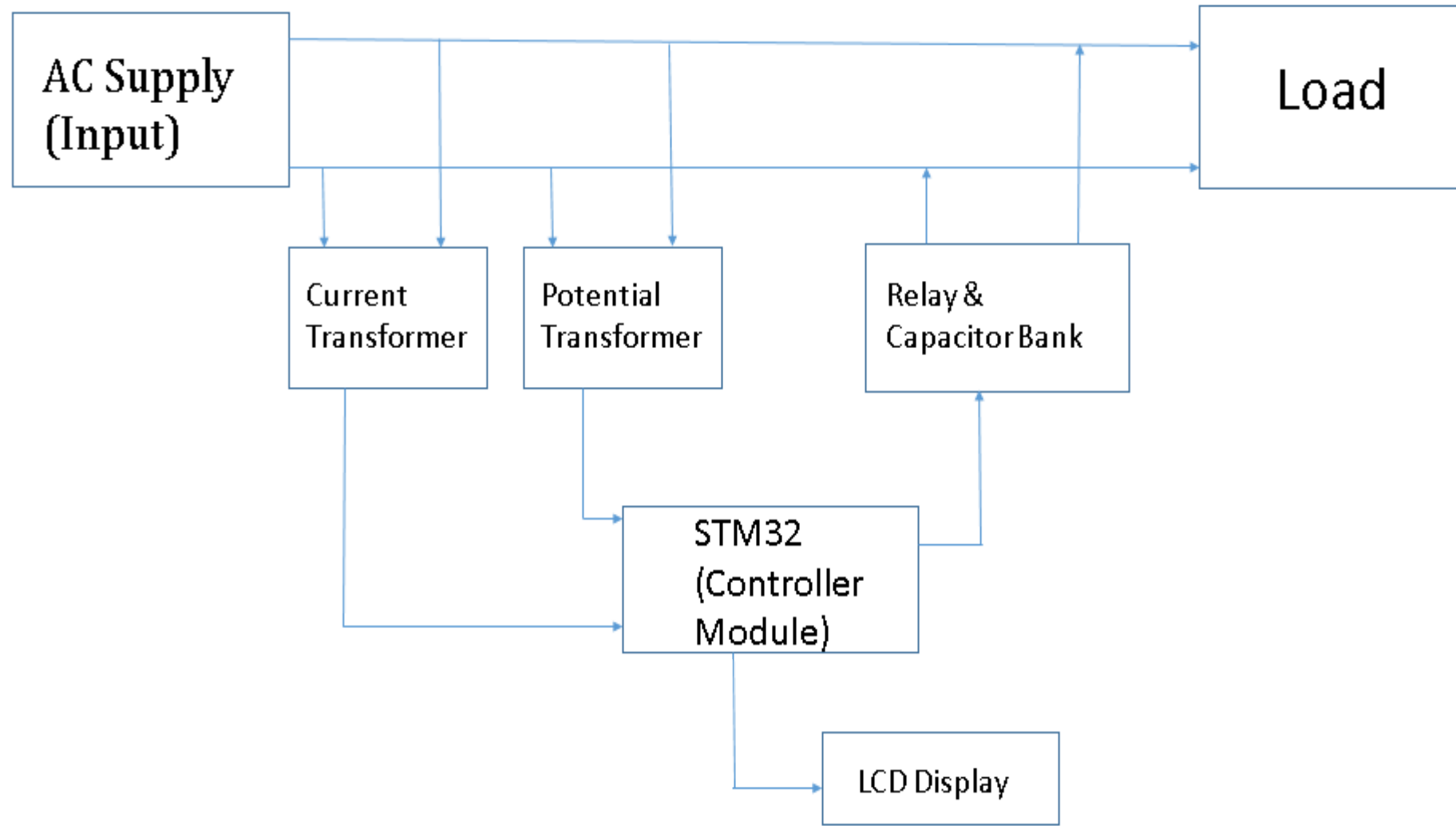


Figure 3.1 – Block Diagram of Automatic Power Factor Control System

Proposed Block Diagram

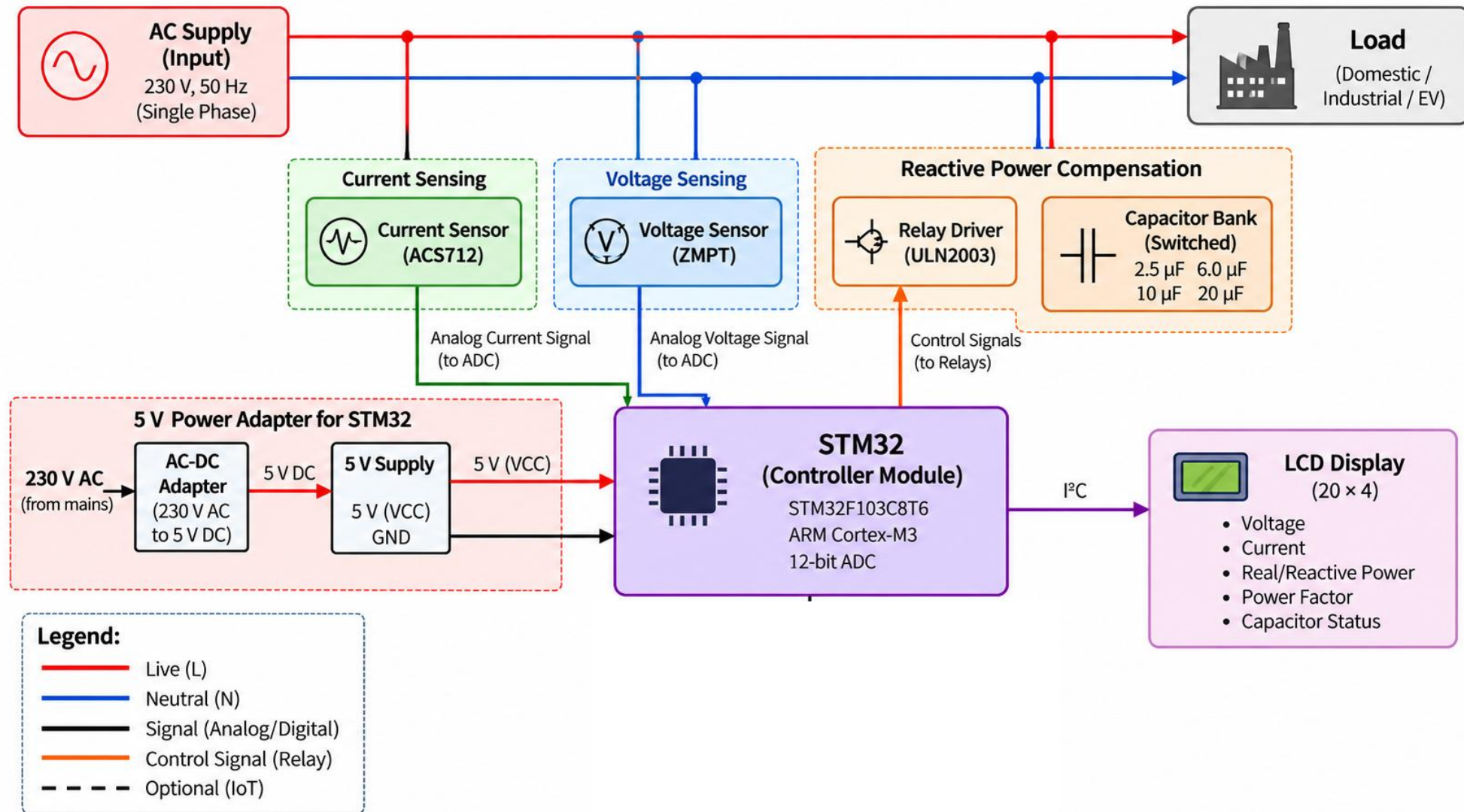


Figure 3.2 Block Diagram

Hardware and Software Requirements

Required Hardware:

- Microcontroller: STM32 Blue Pill (STM32F103C8T6) and ST Link V2
- Power Supply:
 - 230 V/5 V step-down transformer
 - Bridge rectifier
 - Voltage regulators } Power Adapter (5V 1A)
- Voltage Sensing: Potential Transformers (PTs)/ ZMPT101B sensor
- Current Sensing: Current Transformers (CTs)/ ACS712 sensor
- Power Factor Correction: 440 VAC capacitor banks
- Switching Circuit: 4 channel relay driver (with opto coupler)
- Protection: Miniature Circuit Breakers (MCB)
- Display: 20×4 LCD display
- Enclosure: IP54-rated industrial electrical panel

Required Software :

- STM32 Cube ide
- STM32CubeMX
- ST-Link Utility Programmer

Technical Specifications

Parameter	Specification
Input Supply	230 V AC $\pm 10\%$, 50 Hz, single phase
Load	230 V single-phase commercial/inductive load
Main Controller	STM32F103C8T6 Blue Pill
Controller Logic Voltage	3.3 V DC
ADC Resolution	12-bit
ADC Input Range	0–3.3 V
Voltage Sensor	ZMPT voltage sensor
Current Sensor	ACS712 Hall-effect current sensor
Capacitor Stages	2.5 μF , 6 μF , 10 μF , 20 μF
Capacitor Voltage Rating	440 V AC
Switching	4-channel relay module
Display	20 $\times$ 4 Character LCD with I ² C(PC8574)
Protection	MCB

Technical Specifications

Parameter	Specification
Prototype PCB	Zero PCB with soldered header pins
PCB Design Software	KiCad
Firmware Development	STM32CubeIDE
Programmer/Debugger	ST-LINK V2
Specified PF Range	Approximately 0.80–0.99
Maximum Total Capacitance	38.5 μ F
Number of Non-zero Capacitor Combinations	15

RMS Voltage and RMS Current:

$$V_{rms} = \sqrt{\frac{1}{N} \sum_{i=0}^N V[i]^2 - V_{avg}^2}$$

$$I_{rms} = \sqrt{\frac{1}{N} \sum_{i=0}^N i[i]^2 - I_{avg}^2}$$

Real Power:

$$P = \frac{1}{N} \sum_{i=0}^N V[i] \times I[i]$$

Apparent Power:

$$S = V_{rms} \times I_{rms}$$

Capacitor Selection

Phase Angle:

$$\phi_{old} = \cos^{-1}(PF_{old})$$

$$\phi_{target} = \cos^{-1}(PF_{target})$$

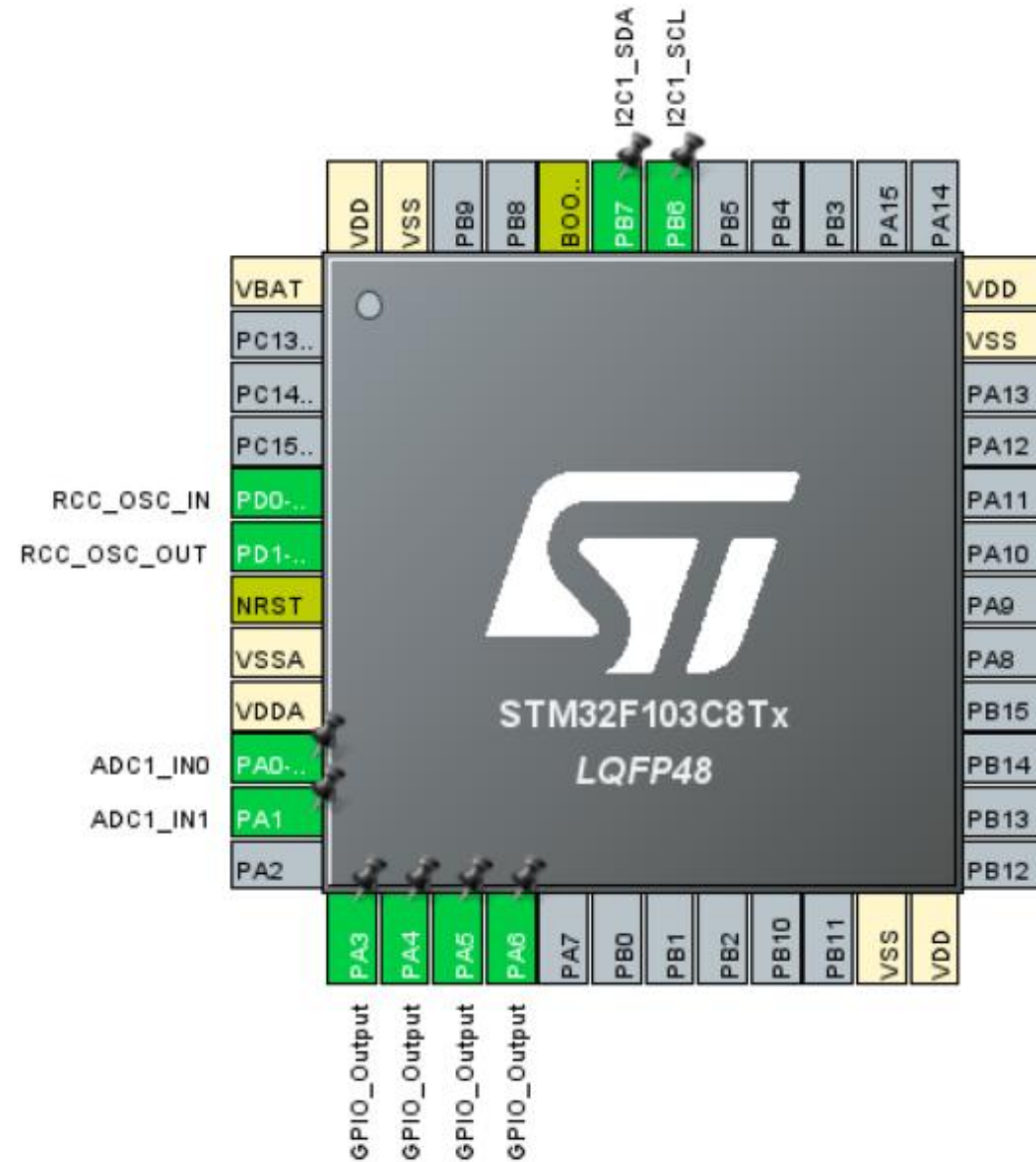
Required reactive compensation:

$$Q_c = P(\tan(\phi_{old}) - \tan(\phi_{target}))$$

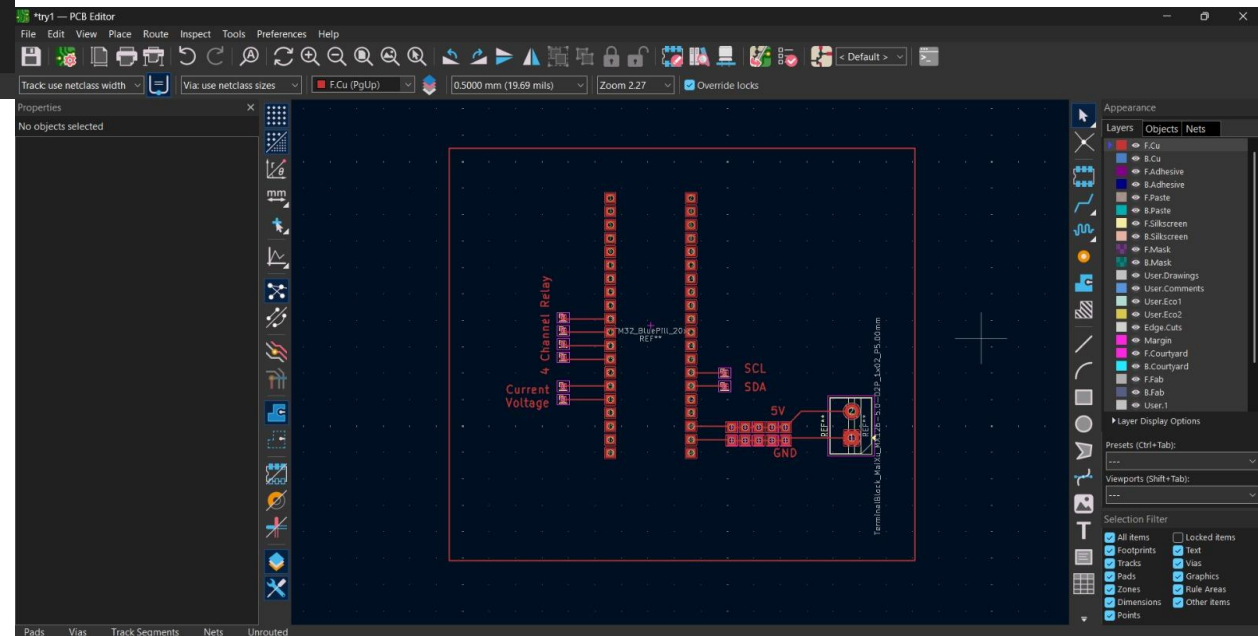
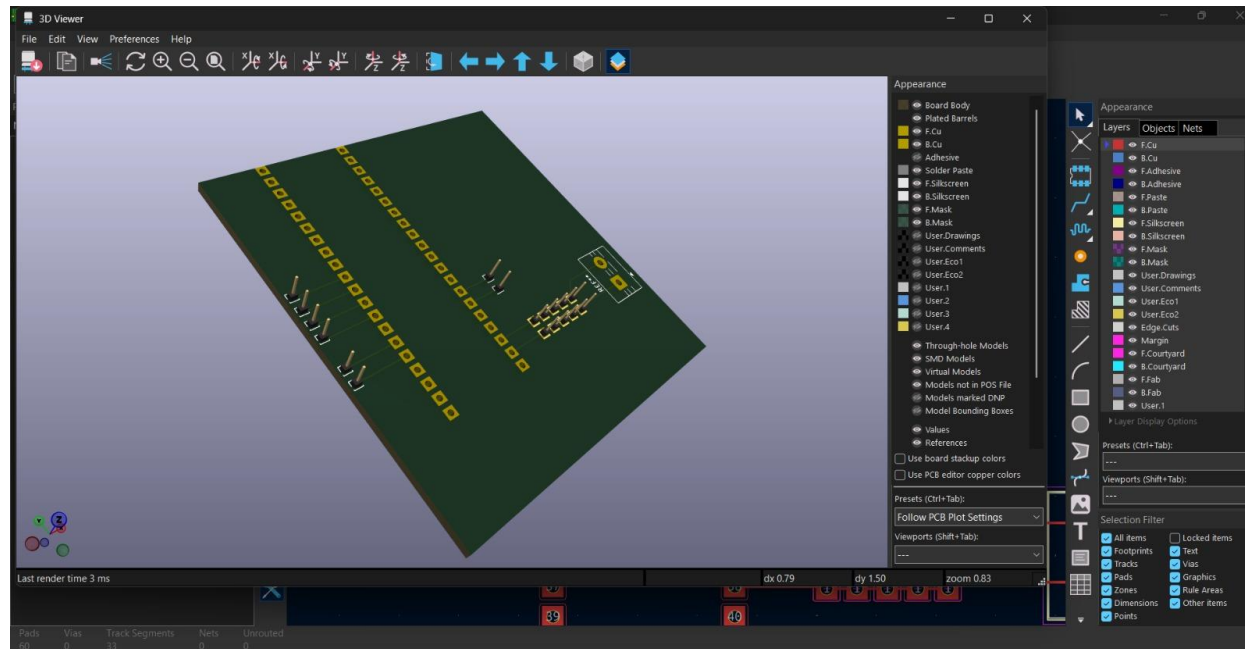
Required Capacitance:

$$C = \frac{Q_c}{2\pi f V_{rms}^2}$$

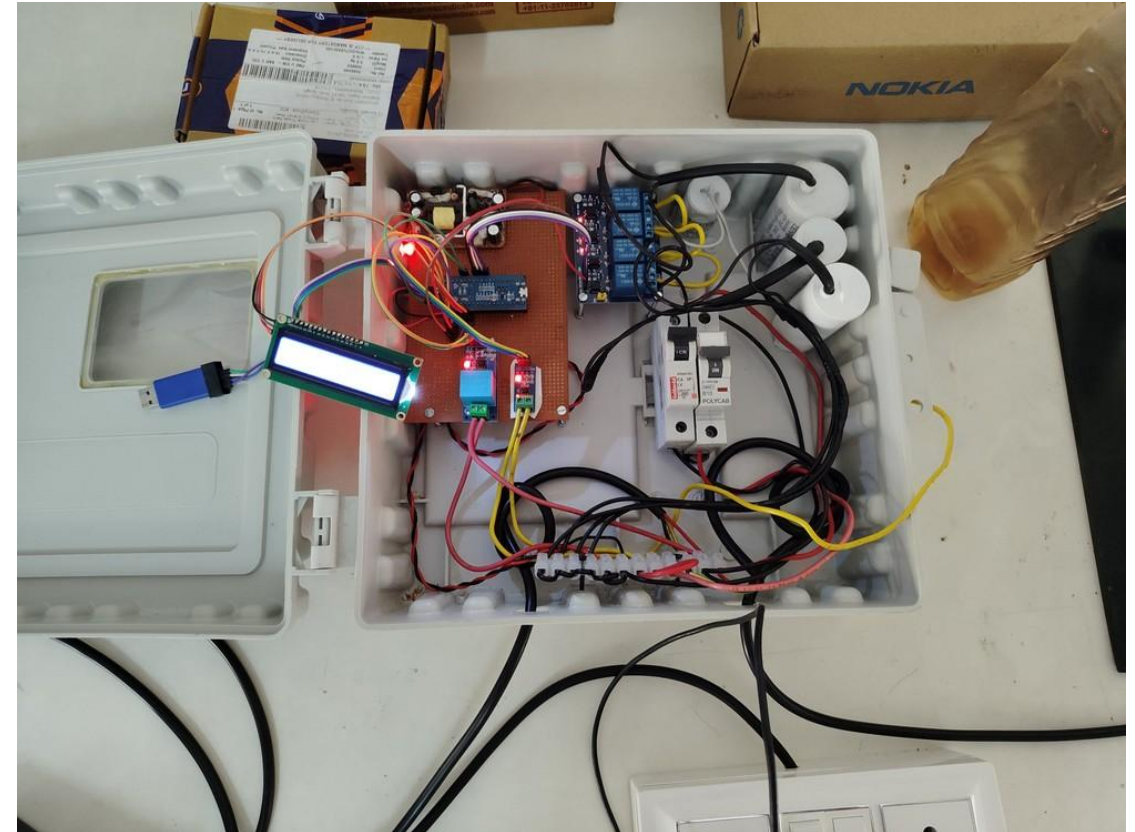
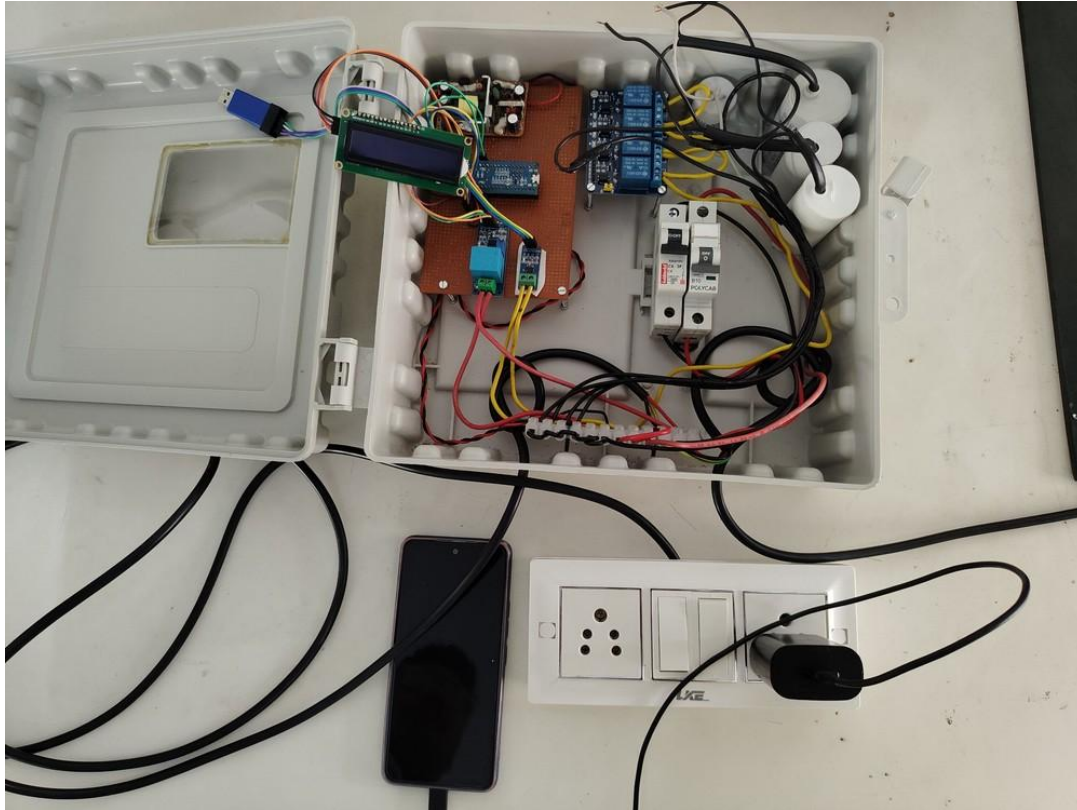
Pin Configuration



PCB



Proposed Solution



Industrial Power Systems:

The proposed system is applicable in manufacturing plants and processing industries for power factor enhancement. It reduces reactive power losses and minimizes demand penalties associated with inductive loads such as motors and drives.

Commercial Buildings:

Applied in commercial complexes with HVAC systems, elevators, and large lighting loads, the system maintains a stable power factor during varying load conditions, leading to improved energy efficiency and reduced operational costs.

High Load Electrical Installations:

Suitable for facilities with varying high electrical loads, ensuring automatic power factor correction, improved power quality, and reduced losses.

Thank You